THE UNIVERSITY OF AUCKLAND

SECOND SEMESTER, 2023
Campus: City

MATHEMATICS

Real Analysis

(Time allowed: TWO hours)

NOTE: Please answer all 8 questions. There are 100 marks available.

The questions are NOT of equal marks.

Show all working and state clearly any theorems that you use in your answers.

Calculators are not permitted in this examination.

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1. (a) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$ and $L \in \mathbb{R}$. Give the definition of $\lim_{n\to\infty} a_n = L$.

(2 marks)

(b) Use the definition of the limit to show that $\lim_{n\to\infty} \dfrac{6n^2}{3n^2 - n - 20} = 2$.

(8 marks)

(c) Prove that every Cauchy sequence is bounded.

(5 marks)

2. (a) Is the series $\sum \dfrac{1+\sqrt{k}}{k^2+\sqrt{k}}$ convergent?

(6 marks)

(b) What is the radius of convergence of the power series $\sum \dfrac{(-1)^k}{2^k + 3^k}\, x^{4k}$?

(7 marks)

3. Let $f\colon \mathbb{R} \to \mathbb{Q}$ be a continuous function. Prove that f is constant.

(6 marks)

4. Let $a < b$ and $f\colon [a,b] \to \mathbb{R}$ be a Riemann integrable function.

(a) Let $m, M \in \mathbb{R}$, and suppose that

$$m \le f(x) \le M, \qquad \text{for all } x \in [a,b].$$

Prove that the "average" of f over $[a,b]$ satisfies

$$m \le \frac{1}{(b-a)} \int_a^b f \le M.$$

(5 marks)

(b) State the mean value theorem.

(3 marks)

(c) Suppose that f is continuous. Prove that there exists a point $c \in [a,b]$ where the average value is taken, i.e.,

$$\frac{1}{(b-a)} \int_a^b f = f(c).$$

Remark: You may not assume that f is differentiable.

(7 marks)

5. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a polynomial with exactly n distinct roots (that is not counting multiplicities).

(a) Show that f' has at least $n-1$ distinct roots.

(5 marks)

(b) Show that $f^{(n-1)}$ has at least one root.

(4 marks)

(c) If f has exactly two distinct roots, then does it follow that f' has exactly one root (not counting multiplicities)?

(4 marks)

6. Let $a < b$ and $f\colon [a,b] \to \mathbb{R}$ be a bounded function. Further let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a,b]$, with

$$a = x_0 < x_1 < \cdots < x_n = b.$$

(a) Define the upper and lower sums $U(f,P)$ and $L(f,P)$ of f for the partition P.

(3 marks)

(b) Suppose $\delta = U(f,P) - L(f,P)$. Prove that

$$L(f,P) \leq \overline{\int_a^b} f(x)\,dx \leq L(f,P) + \delta.$$

(4 marks)

Fix a point $c \in (a,b)$, and $y_1, y_2 \in \mathbb{R}$ with $y_1 \leq y_2$. Define $g\colon [a,b] \to \mathbb{R}$ by

$$g(x) := \begin{cases} y_1, & \text{if } a \leq x \leq c; \\ y_2, & \text{if } c < x \leq b. \end{cases}$$

(c) Let $k \in \{1, \ldots, n\}$ and suppose that $c \in (x_{k-1}, x_k)$. Compute $U(g,P)$ and $L(g,P)$ explicitly.

(4 marks)

(d) Verify that g is Riemann integrable by checking the Riemann condition.

(4 marks)

(e) Use part 6(c) to determine the Riemann integral of g.

(4 marks)

7. Explain why the derivative

$$\frac{d}{dx}\Big(\int_x^{x^2} e^{t^2}\,dt\Big),$$

exists, and calculate it, explaining what tools/results you use.

Hint: Write the integral in terms of $F(y) = \displaystyle\int_0^y e^{t^2}\,dt$, but do not try to evaluate F.

(8 marks)

8. For all $n \in \mathbb{N}$ define the continuous function $f_n \colon [0,1] \to \mathbb{R}$ by

$$f_n(x) := \frac{nx}{nx+1}.$$

(a) For all $x \in [0,1]$, calculate $\lim_{n\to\infty} f_n(x)$.

(4 marks)

(b) Is the sequence $(f_n)_{n\in\mathbb{N}}$ uniformly convergent?
Hint: Use part (a).

(7 marks)